

B6: Condensed-Matter Physics — 2024 Model Solutions

Trinity Term 2024 (Paper A13485W1)

Question 1 — Sodium under pressure: bcc–fcc transition and X-ray diffraction

Problem. Na is bcc at $p = 0$ ($a_0 = 4.27 \text{ \AA}$) and undergoes a first-order bcc→fcc transition at $p_c = 65 \text{ GPa}$, with $a = 2.90 \text{ \AA}$ just below and $a' = 3.65 \text{ \AA}$ just above. Powder XRD uses $\lambda = 0.43 \text{ \AA}$; molar mass $M = 23 \text{ g mol}^{-1}$.

(a)(i)–(ii) Conventional unit cells and atoms per cell



Count bcc corners (each shared $\times 1/8$) plus one body-centre:

$$n_{\text{bcc}} = 8 \times \frac{1}{8} + 1.$$

Add the two contributions:

$$n_{\text{bcc}} = 1 + 1 = 2.$$

Count fcc corners plus six face-centres (each shared $\times 1/2$):

$$n_{\text{fcc}} = 8 \times \frac{1}{8} + 6 \times \frac{1}{2}.$$

Add the two contributions:

$$n_{\text{fcc}} = 1 + 3 = 4.$$

$n_{\text{bcc}} = 2, \quad n_{\text{fcc}} = 4.$

(a)(iii)–(iv) The (111) plane and nearest-neighbour distance in it

bcc (111) contains only the three corners; the body-centre lies off-plane. Triangle side equals the face diagonal:

$$d_{\text{bcc}}^{(111)} = \sqrt{2} a.$$

Plug in $a = 2.90 \text{ \AA}$:

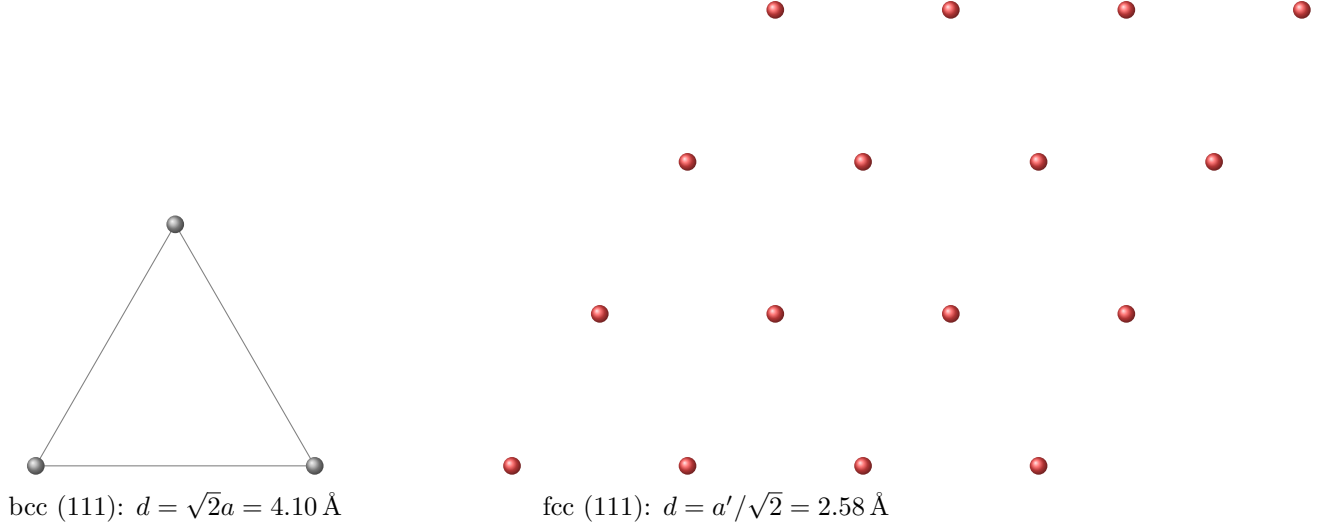
$$d_{\text{bcc}}^{(111)} = \sqrt{2} \times 2.90 \text{ \AA} = 4.10 \text{ \AA}.$$

fcc (111) is close-packed; nearest-neighbour separation equals half the face diagonal:

$$d_{\text{fcc}}^{(111)} = \frac{a'}{\sqrt{2}}.$$

Plug in $a' = 3.65 \text{ \AA}$:

$$d_{\text{fcc}}^{(111)} = 3.65/\sqrt{2} \text{ \AA} = 2.58 \text{ \AA}.$$



$d_{\text{bcc}}^{(111)} = 4.10 \text{ \AA}, \quad d_{\text{fcc}}^{(111)} = 2.58 \text{ \AA}.$

(b)(i) Diffraction selection rules and reflection positions

Bragg + cubic spacing $d_{hkl} = a/\sqrt{h^2 + k^2 + \ell^2}$:

$$\sin \theta = \frac{\lambda}{2a} \sqrt{N}, \quad N \equiv h^2 + k^2 + \ell^2. \quad (1)$$

Selection rules:

bcc: $h + k + \ell$ even; fcc: h, k, ℓ all even or all odd.

(2)

Allowed N sequences:

$$N_{\text{bcc}} : 2, 4, 6, 8, \dots \quad (hkl) = 110, 200, 211, 220, \dots$$

$$N_{\text{fcc}} : 3, 4, 8, 11, 12, \dots \quad (hkl) = 111, 200, 220, 311, 222, \dots$$

Window $2\theta < 25^\circ$ requires $\sin \theta < 0.2164$.

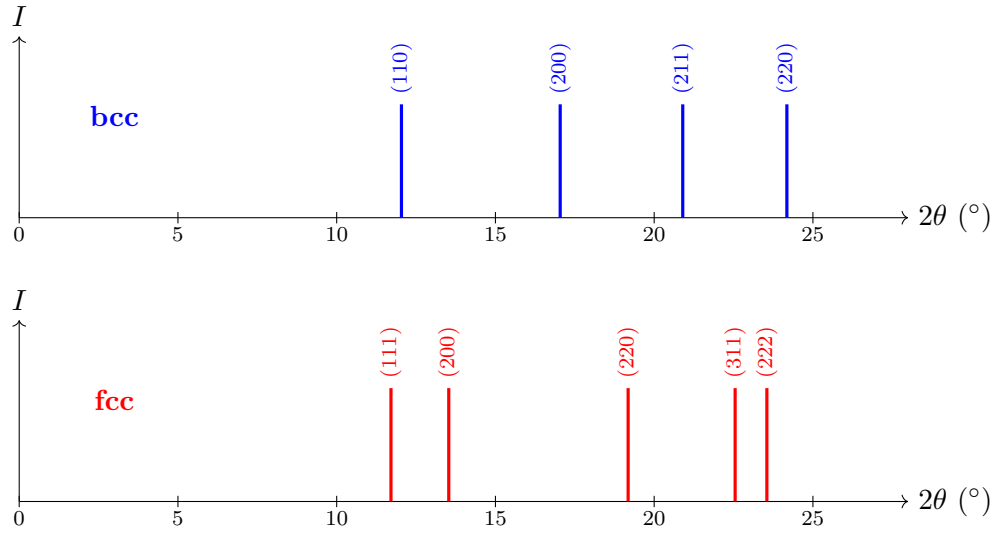
bcc, $\lambda/(2a) = 0.07414$, $\sqrt{N} < 2.919$, $N \leq 8$:

$$\begin{aligned}
 (110) : N = 2, \quad 2\theta &= 12.04^\circ; \\
 (200) : N = 4, \quad 2\theta &= 17.04^\circ; \\
 (211) : N = 6, \quad 2\theta &= 20.90^\circ; \\
 (220) : N = 8, \quad 2\theta &= 24.18^\circ.
 \end{aligned} \quad (3)$$

fcc, $\lambda/(2a') = 0.05890$, $\sqrt{N} < 3.674$, $N \leq 12$:

$$\begin{aligned}
 (111) : N = 3, \quad 2\theta &= 11.71^\circ; \\
 (200) : N = 4, \quad 2\theta &= 13.53^\circ; \\
 (220) : N = 8, \quad 2\theta &= 19.18^\circ; \\
 (311) : N = 11, \quad 2\theta &= 22.55^\circ; \\
 (222) : N = 12, \quad 2\theta &= 23.55^\circ.
 \end{aligned} \quad (4)$$

(b)(ii) Sketch of the two diffraction patterns



(b)(iii) Lattice parameter at $p_1 = 100$ GPa

Third allowed fcc reflection is (220) with $N = 8$. Shift up by 1.20° :

$$2\theta_{220}^{(1)} = 19.18 + 1.20 = 20.38^\circ.$$

Halve to get θ :

$$\theta_{220}^{(1)} = 10.19^\circ.$$

Evaluate the sine:

$$\sin \theta_{220}^{(1)} = \sin(10.19^\circ) = 0.17688.$$

Apply (1) with $\sqrt{N} = 2\sqrt{2}$:

$$a'_1 = \frac{\lambda \sqrt{N}}{2 \sin \theta_{220}^{(1)}}.$$

Substitute the numbers:

$$a'_1 = \frac{0.43 \times 2\sqrt{2}}{2 \times 0.17688} \text{ \AA}.$$

Reduce the numerator and denominator:

$$a'_1 = \frac{1.2162}{0.35376} \text{ \AA} = 3.438 \text{ \AA}.$$

$$\boxed{a'_1 = 3.44 \text{ \AA}.} \quad (5)$$

(c) Hard-sphere atomic radius and density change at p_c

bcc spheres touch along the body diagonal $\sqrt{3}a = 4r$:

$$r_{\text{bcc}} = \frac{\sqrt{3}}{4}a.$$

Plug in $a = 2.90 \text{ \AA}$:

$$r_{\text{bcc}} = \frac{\sqrt{3}}{4} \times 2.90 \text{ \AA} = 1.256 \text{ \AA}.$$

fcc spheres touch along the face diagonal $\sqrt{2}a' = 4r$:

$$r_{\text{fcc}} = \frac{\sqrt{2}}{4}a'.$$

Plug in $a' = 3.65 \text{ \AA}$:

$$r_{\text{fcc}} = \frac{\sqrt{2}}{4} \times 3.65 \text{ \AA} = 1.291 \text{ \AA}.$$

Number density $n = n_{\text{cell}}/a^3$, bcc:

$$n_b = \frac{2}{(2.90 \times 10^{-10})^3} \text{ m}^{-3}.$$

Evaluate:

$$n_b = \frac{2}{24.389 \times 10^{-30}} \text{ m}^{-3} = 8.20 \times 10^{28} \text{ m}^{-3}.$$

Same for fcc:

$$n_f = \frac{4}{(3.65 \times 10^{-10})^3} \text{ m}^{-3}.$$

Evaluate:

$$n_f = \frac{4}{48.627 \times 10^{-30}} \text{ m}^{-3} = 8.226 \times 10^{28} \text{ m}^{-3}.$$

Mass density $\rho = nM/N_A$, bcc:

$$\rho_{\text{bcc}} = \frac{8.20 \times 10^{28} \times 23 \times 10^{-3}}{6.022 \times 10^{23}} \text{ kg m}^{-3}.$$

Reduce:

$$\rho_{\text{bcc}} = 3131 \text{ kg m}^{-3}.$$

fcc:

$$\rho_{\text{fcc}} = \frac{8.226 \times 10^{28} \times 23 \times 10^{-3}}{6.022 \times 10^{23}} \text{ kg m}^{-3}.$$

Reduce:

$$\rho_{\text{fcc}} = 3141 \text{ kg m}^{-3}.$$

Take the difference:

$$\Delta\rho = \rho_{\text{fcc}} - \rho_{\text{bcc}} = +10 \text{ kg m}^{-3}.$$

Form the fractional change:

$$\Delta\rho/\rho_{\text{bcc}} = 10/3131 = 0.32\%.$$

$$\boxed{\Delta\rho/\rho_{\text{bcc}} = +0.32\%, \quad \Delta\rho \approx +10 \text{ kg m}^{-3}.}$$

(6)

(d) Compressibility of the bcc phase between ambient and p_c

Treat $\beta = -(1/V)\partial V/\partial p$ as constant and integrate:

$$V(p) = V_0 e^{-\beta p}.$$

Solve for β at $p = p_c$:

$$\beta = -\frac{1}{p_c} \ln \frac{V}{V_0}.$$

Use $V \propto a^3$:

$$\beta = -\frac{3}{p_c} \ln \frac{a}{a_0}.$$

Evaluate the log:

$$\ln(2.90/4.27) = -0.38670.$$

Plug in numbers with $p_c = 65 \times 10^9 \text{ Pa}$:

$$\beta_{\text{bcc}} = \frac{3 \times 0.38670}{6.5 \times 10^{10}} \text{ Pa}^{-1}.$$

Reduce:

$$\beta_{\text{bcc}} = \frac{1.1601}{6.5 \times 10^{10}} \text{ Pa}^{-1} = 1.785 \times 10^{-11} \text{ Pa}^{-1}.$$

$$\boxed{\beta_{\text{bcc}} = 1.78 \times 10^{-11} \text{ Pa}^{-1}.} \quad (7)$$

(e) Mass of sodium that fits in the pressure cell

Cylindrical volume with radius $50 \mu\text{m}$ and height 1 mm :

$$V_{\text{cyl}} = \pi(50 \times 10^{-6})^2 \times 10^{-3} \text{ m}^3.$$

Evaluate:

$$V_{\text{cyl}} = \pi \times 2.5 \times 10^{-9} \times 10^{-3} \text{ m}^3 = 7.854 \times 10^{-12} \text{ m}^3.$$

With 35% void, Na occupies $0.65 V_{\text{cyl}}$:

$$V_{\text{Na}} = 0.65 \times 7.854 \times 10^{-12} \text{ m}^3 = 5.105 \times 10^{-12} \text{ m}^3.$$

Ambient bcc density uses $a_0^3 = (4.27 \times 10^{-10})^3 = 77.854 \times 10^{-30} \text{ m}^3$:

$$\rho_0 = \frac{2 \times 23 \times 10^{-3}}{6.022 \times 10^{23} \times 77.854 \times 10^{-30}} \text{ kg m}^{-3}.$$

Reduce:

$$\rho_0 = \frac{46 \times 10^{-3}}{4.689 \times 10^{-5}} \text{ kg m}^{-3} = 981 \text{ kg m}^{-3}.$$

Mass:

$$m_{\text{Na}} = \rho_0 V_{\text{Na}} = 981 \times 5.105 \times 10^{-12} \text{ kg}.$$

Reduce:

$$m_{\text{Na}} = 5.008 \times 10^{-9} \text{ kg}.$$

$$\boxed{m_{\text{Na}} \approx 5.0 \times 10^{-9} \text{ kg} \approx 5 \text{ ng}.} \quad (8)$$

Question 2 — Heat capacity of YbMg_2Bi_2 in 3D and 2D

Problem. 3D molar heat capacity is $C = zN_A k_B [(\pi^2/2)(T/T_F) + (12\pi^4/5)(T/T_D)^3]$ at low T , saturating at high T . (a) Identify z from the saturation. (b) Justify each term, extract T_F, T_D from C/T vs T^2 , find cross-over T^* . (c) For a 2D film with two in-plane polarisations, find $g_{\text{ph}}^{(2)}(\omega)$, the high- T limit, and the low- T form $C = AT + BT^\delta$ with $A(g(E_F))$, $B(\omega_D)$. Use $\int_0^\infty x^3/(e^x - 1) dx = \pi^4/15$, $\int_0^\infty x^2/(e^x - 1) dx = 2\zeta(3) \simeq 2.4$.

(a) Origin of the high- T saturation and Dulong–Petit limit

Equipartition: each of $3N$ classical modes contributes k_B :

$$C_{\text{lat}}^{(\infty)} = 3Nk_B.$$

Per mole of formula units use $N = zN_A$:

$$C^{(\infty)} = 3zN_Ak_B = 3zR.$$

Read $C^{(\infty)} \simeq 120 \text{ J mol}^{-1} \text{ K}^{-1}$ from the figure:

$$z = \frac{120}{3R} = \frac{120}{3 \times 8.314}.$$

Reduce:

$$z \approx 4.81.$$

$$\boxed{z = 5 \text{ atoms per formula unit.}} \quad (9)$$

(b)(i) Justification of the two terms

Linear electronic term. Sommerfeld expansion to leading order in T/T_F :

$$U_{\text{el}}(T) = U_{\text{el}}(0) + \frac{\pi^2}{6} g(E_F) (k_B T)^2. \quad (10)$$

Differentiate:

$$C_{\text{el}} = \frac{\pi^2}{3} g(E_F) k_B^2 T \equiv \gamma T. \quad (11)$$

Free-electron form $g(E_F) = 3N_e/(2k_B T_F)$. Substitute:

$$C_{\text{el}} = \frac{\pi^2}{3} \cdot \frac{3N_e}{2k_B T_F} k_B^2 T.$$

Cancel and tidy:

$$C_{\text{el}} = \frac{\pi^2}{2} N_e k_B \frac{T}{T_F}.$$

Identify $N_e = zN_A$ to recover the stated prefactor.

Cubic Debye term. 3D phonon DOS (3 acoustic branches):

$$g_{\text{ph}}(\omega) = \frac{3V\omega^2}{2\pi^2 v_s^3}. \quad (12)$$

Internal energy is a Bose integral over the DOS:

$$U_{\text{ph}}(T) = \int_0^{\omega_D} \frac{\hbar\omega}{e^{\hbar\omega/k_B T} - 1} g_{\text{ph}}(\omega) d\omega.$$

Substitute $x = \hbar\omega/k_B T$:

$$U_{\text{ph}}(T) = \frac{3V}{2\pi^2 v_s^3 \hbar^3} (k_B T)^4 \int_0^{x_D} \frac{x^3}{e^x - 1} dx.$$

Low T , extend the upper limit to ∞ :

$$\int_0^{\infty} \frac{x^3}{e^x - 1} dx = \frac{\pi^4}{15}.$$

Insert this value:

$$U_{\text{ph}}(T) \approx \frac{3V}{2\pi^2 v_s^3 \hbar^3} (k_{\text{B}} T)^4 \cdot \frac{\pi^4}{15}.$$

Eliminate V/v_s^3 via the Debye condition $\omega_D^3 = 6\pi^2 N v_s^3/V$:

$$\frac{V}{v_s^3} = \frac{6\pi^2 N}{\omega_D^3}.$$

Use $\hbar\omega_D = k_{\text{B}}T_D$:

$$\frac{V}{v_s^3} = \frac{6\pi^2 N \hbar^3}{(k_{\text{B}} T_D)^3}.$$

Substitute into U_{ph} :

$$U_{\text{ph}}(T) = \frac{3(k_{\text{B}} T)^4}{2\pi^2 \hbar^3} \cdot \frac{6\pi^2 N \hbar^3}{(k_{\text{B}} T_D)^3} \cdot \frac{\pi^4}{15}.$$

Cancel $\hbar^3$ and $2\pi^2$:

$$U_{\text{ph}}(T) = \frac{3\pi^4}{5} N k_{\text{B}} \frac{T^4}{T_D^3}.$$

Differentiate with respect to T :

$$C_{\text{ph}}(T) = \frac{12\pi^4}{5} N k_{\text{B}} \frac{T^3}{T_D^3}.$$

Per mole use $N = zN_{\text{A}}$:

$$\boxed{C_{\text{ph}}(T) = \frac{12\pi^4}{5} z N_{\text{A}} k_{\text{B}} \frac{T^3}{T_D^3}.} \quad (13)$$

(b)(ii) Reading T_{F} and T_{D} from the C/T vs T^2 plot

Model:

$$\frac{C}{T} = \gamma + \beta T^2, \quad \gamma = \frac{\pi^2 z R}{2T_{\text{F}}}, \quad \beta = \frac{12\pi^4 z R}{5T_{\text{D}}^3}. \quad (14)$$

Read intercept $\gamma \simeq 0.30 \text{ J mol}^{-1} \text{ K}^{-2}$, slope $\beta \simeq 1.6 \times 10^{-3} \text{ J mol}^{-1} \text{ K}^{-4}$.

Solve for T_{F} :

$$T_{\text{F}} = \frac{\pi^2 z R}{2\gamma}.$$

Plug in numbers:

$$T_{\text{F}} = \frac{\pi^2 \times 5 \times 8.314}{2 \times 0.30} \text{ K}.$$

Reduce:

$$T_{\text{F}} = \frac{410.3}{0.60} \text{ K} \approx 684 \text{ K}.$$

$$\boxed{T_{\text{F}} \approx 7 \times 10^2 \text{ K}.} \quad (15)$$

Solve for T_{D}^3 :

$$T_{\text{D}}^3 = \frac{12\pi^4 z R}{5\beta}.$$

Plug in numbers:

$$T_{\text{D}}^3 = \frac{12\pi^4 \times 5 \times 8.314}{5 \times 1.6 \times 10^{-3}} \text{ K}^3.$$

Reduce numerator $12\pi^4 \times 5 \times 8.314 = 4.86 \times 10^4$:

$$T_D^3 = \frac{4.86 \times 10^4}{8.0 \times 10^{-3}} \text{ K}^3 = 6.07 \times 10^6 \text{ K}^3.$$

Cube root:

$$T_D \approx 183 \text{ K}.$$

$$\boxed{T_D \approx 2 \times 10^2 \text{ K}.} \quad (16)$$

(b)(iii) Cross-over temperature T^*

Equate the two contributions:

$$\frac{\pi^2}{2} \frac{T^*}{T_F} = \frac{12\pi^4}{5} \left(\frac{T^*}{T_D} \right)^3.$$

Cancel one factor of T^* and rearrange:

$$1 = \frac{24\pi^2 (T^*)^2 T_F}{5T_D^3}.$$

Solve for $(T^*)^2$:

$$(T^*)^2 = \frac{5T_D^3}{24\pi^2 T_F}.$$

Plug in $T_D = 183 \text{ K}$, $T_F = 684 \text{ K}$:

$$(T^*)^2 = \frac{5 \times (183)^3}{24\pi^2 \times 684} \text{ K}^2.$$

Reduce numerator $5 \times 183^3 = 3.064 \times 10^7$:

$$(T^*)^2 = \frac{3.064 \times 10^7}{1.620 \times 10^5} \text{ K}^2 = 189 \text{ K}^2.$$

Take the square root:

$$T^* \approx 13.7 \text{ K}.$$

$$\boxed{T^* \approx 14 \text{ K}.} \quad (17)$$

(c)(i) 2D phonon density of states

Two in-plane polarisations, density $A/(2\pi)^2$ per polarisation in $\mathbf{q}$ -space:

$$g_{\text{ph}}^{(2)}(\omega) d\omega = 2 \cdot \frac{A}{(2\pi)^2} \cdot 2\pi q dq.$$

Use linear dispersion $\omega = v_s q$ so $q = \omega/v_s$ and $dq = d\omega/v_s$:

$$g_{\text{ph}}^{(2)}(\omega) d\omega = \frac{A}{\pi} \frac{\omega}{v_s} \frac{d\omega}{v_s}.$$

Combine the v_s factors:

$$g_{\text{ph}}^{(2)}(\omega) d\omega = \frac{A\omega}{\pi v_s^2} d\omega.$$

$$\boxed{g_{\text{ph}}^{(2)}(\omega) = \frac{A\omega}{\pi v_s^2}.} \quad (18)$$

(c)(ii) High- T Dulong–Petit in 2D

Total in-plane modes = $2N$, equipartition gives k_B each:

$$C_{2D}^{(\infty)} = 2Nk_B.$$

Per mole use $N = zN_A$ and $z = 5$:

$$C_{\text{molar}}^{(\infty)} = 2zR = 10R.$$

Plug in $R = 8.314$:

$$C_{\text{molar}}^{(\infty)} = 10 \times 8.314 \approx 83 \text{ J mol}^{-1} \text{ K}^{-1}.$$

$$\boxed{C_{\text{molar}}^{(\infty)} = 2zR = 10R \approx 83 \text{ J mol}^{-1} \text{ K}^{-1}.} \quad (19)$$

(c)(iii) Low- T form $C = AT + BT^\delta$ in 2D

Phonon energy with 2D DOS:

$$U_{\text{ph}}^{(2)}(T) = \int_0^{\omega_D} \frac{\hbar\omega}{e^{\hbar\omega/k_B T} - 1} \frac{A\omega}{\pi v_s^2} d\omega.$$

Substitute $x = \hbar\omega/k_B T$, so $\omega = k_B T x / \hbar$ and $d\omega = k_B T dx / \hbar$:

$$U_{\text{ph}}^{(2)}(T) = \frac{A}{\pi v_s^2 \hbar^2} (k_B T)^3 \int_0^{x_D} \frac{x^2}{e^x - 1} dx.$$

Low T , extend the upper limit to ∞ :

$$\int_0^\infty \frac{x^2}{e^x - 1} dx = 2\zeta(3).$$

Insert this value:

$$U_{\text{ph}}^{(2)}(T) \approx \frac{2\zeta(3) A}{\pi v_s^2 \hbar^2} (k_B T)^3.$$

Differentiate with respect to T :

$$C_{\text{ph}}^{(2)}(T) = \frac{6\zeta(3) A k_B^3}{\pi v_s^2 \hbar^2} T^2.$$

$$\boxed{C_{\text{ph}}^{(2)}(T) = \frac{6\zeta(3) A k_B^3}{\pi v_s^2 \hbar^2} T^2 \equiv BT^\delta, \quad \delta = 2.} \quad (20)$$

(c)(iv) Identification of A in terms of $g(E_F)$, and B in terms of ω_D

Sommerfeld result (11) is dimension-independent:

$$\boxed{A = \frac{\pi^2}{3} k_B^2 g(E_F).} \quad (21)$$

Debye condition: total in-plane modes equals $2N$:

$$2N = \int_0^{\omega_D} \frac{A\omega}{\pi v_s^2} d\omega.$$

Integrate:

$$2N = \frac{A\omega_D^2}{2\pi v_s^2}.$$

Rearrange:

$$\frac{A}{\pi v_s^2} = \frac{4N}{\omega_D^2}.$$

Substitute into the result for $C_{\text{ph}}^{(2)}$:

$$C_{\text{ph}}^{(2)} = \frac{6\zeta(3)k_{\text{B}}^3}{\hbar^2} \cdot \frac{4N}{\omega_D^2} T^2.$$

Combine factors:

$$C_{\text{ph}}^{(2)} = \frac{24\zeta(3) N k_{\text{B}}^3}{\hbar^2 \omega_D^2} T^2.$$

Per mole use $N = zN_{\text{A}}$:

$$\boxed{B = \frac{24\zeta(3) z N_{\text{A}} k_{\text{B}}^3}{\hbar^2 \omega_D^2}.} \quad (22)$$

Question 3 — Tight-binding band on a 2D square lattice

Problem. 2D square lattice (parameter a), tight-binding $\epsilon(k_x, k_y) = \epsilon_0 + 2t \cos k_x a + 2t \cos k_y a$, $t > 0$. Find dispersion along $\Gamma \rightarrow X \rightarrow M$, bandwidth, contours, effective masses; the linear $\epsilon_{\text{F}} = \epsilon'_{\text{F}} + \alpha N/A$ near the band minimum and $g(\epsilon) = \beta A$; for $A_{\text{FS}}/A_{\text{BZ}} = 1/100$ at $a = 3.5 \text{ \AA}$, find n_{2D} and Hall voltage.

(a)(i) Band dispersion along $\Gamma \rightarrow X \rightarrow M$

Evaluate at $\Gamma = (0, 0)$:

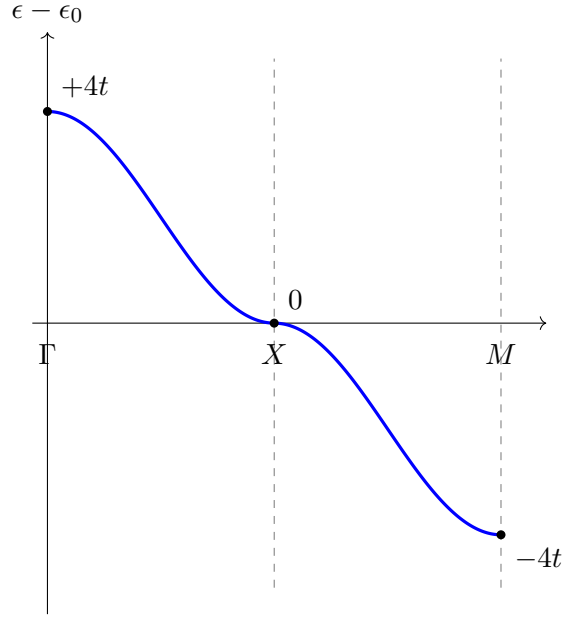
$$\epsilon(\Gamma) = \epsilon_0 + 2t \cos 0 + 2t \cos 0 = \epsilon_0 + 4t.$$

Evaluate at $X = (\pi/a, 0)$:

$$\epsilon(X) = \epsilon_0 + 2t \cos \pi + 2t \cos 0 = \epsilon_0.$$

Evaluate at $M = (\pi/a, \pi/a)$:

$$\epsilon(M) = \epsilon_0 + 2t \cos \pi + 2t \cos \pi = \epsilon_0 - 4t.$$



(a)(ii) Bandwidth

$$W = \epsilon_{\max} - \epsilon_{\min} = 8t. \quad (23)$$

(a)(iii) Constant-energy contours

Expand around $M = (\pi/a, \pi/a)$, $\mathbf{k} = \mathbf{K}_M + \mathbf{q}$, using $\cos(\pi + qa) \simeq -1 + (qa)^2/2$:

$$\epsilon(\mathbf{k}) \simeq \epsilon_0 + 2t[-1 + (q_x a)^2/2] + 2t[-1 + (q_y a)^2/2].$$

Collect terms:

$$\epsilon(\mathbf{k}) \simeq \epsilon_0 - 4t + ta^2(q_x^2 + q_y^2). \quad (24)$$

Expand around Γ , using $\cos(qa) \simeq 1 - (qa)^2/2$:

$$\epsilon(\mathbf{k}) \simeq \epsilon_0 + 2t[1 - (q_x a)^2/2] + 2t[1 - (q_y a)^2/2].$$

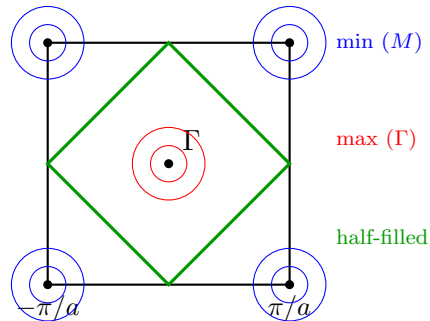
Collect terms:

$$\epsilon(\mathbf{k}) \simeq \epsilon_0 + 4t - ta^2(q_x^2 + q_y^2). \quad (25)$$

Both contours are circular near the extrema. Half-filling: $\epsilon = \epsilon_0$ gives $\cos k_x a = -\cos k_y a$:

$$k_x a = \pm(\pi - k_y a).$$

The contour is a square rotated 45° joining the four X -points.



(a)(iv) Effective masses near band edges

Use $1/m^* = \hbar^{-2} \partial^2 \epsilon / \partial q^2$. From (24):

$$\frac{\partial^2 \epsilon}{\partial q_x^2} = 2ta^2.$$

Invert:

$$m_{\min}^* = +\frac{\hbar^2}{2ta^2} > 0 \quad (\text{electron-like}).$$

From (25):

$$\frac{\partial^2 \epsilon}{\partial q_x^2} = -2ta^2.$$

Invert:

$$m_{\max}^* = -\frac{\hbar^2}{2ta^2} < 0 \quad (\text{hole-like}).$$

$$\boxed{m_{\min}^* = +\frac{\hbar^2}{2ta^2}, \quad m_{\max}^* = -\frac{\hbar^2}{2ta^2}.} \quad (26)$$

(b)(i)–(ii) $\epsilon_F(N/A)$ near the band minimum

Parabolic band near M with $m^* = \hbar^2/(2ta^2)$. Count states inside the Fermi disc, factor 2 for spin:

$$N(\epsilon_F) = 2 \cdot \frac{A}{(2\pi)^2} \cdot \pi q_F^2.$$

Simplify:

$$N(\epsilon_F) = \frac{A q_F^2}{2\pi}.$$

From (24) relate q_F to ϵ_F :

$$ta^2 q_F^2 = \epsilon_F - (\epsilon_0 - 4t).$$

Solve for q_F^2 :

$$q_F^2 = \frac{\epsilon_F - \epsilon_0 + 4t}{ta^2}.$$

Substitute into N/A :

$$\frac{N}{A} = \frac{\epsilon_F - \epsilon_0 + 4t}{2\pi ta^2}.$$

Solve for ϵ_F :

$$\epsilon_F = (\epsilon_0 - 4t) + 2\pi ta^2 \frac{N}{A}.$$

$$\boxed{\epsilon_F = \epsilon'_F + \alpha \frac{N}{A}, \quad \epsilon'_F = \epsilon_0 - 4t, \quad \alpha = 2\pi ta^2.} \quad (27)$$

(b)(iii) Density of states near the band minimum

Differentiate N with respect to ϵ_F :

$$g(\epsilon) = \frac{\partial N}{\partial \epsilon}.$$

Apply to $N = A(\epsilon_F - \epsilon_0 + 4t)/(2\pi ta^2)$:

$$g(\epsilon) = \frac{A}{2\pi ta^2}.$$

Identify β via $g = \beta A$:

$$\boxed{\beta = \frac{1}{2\pi ta^2}.} \quad (28)$$

(c)(i) Carrier density for $A_{\text{FS}}/A_{\text{BZ}} = 1/100$

BZ area $A_{\text{BZ}} = (2\pi/a)^2$. Spin factor 2 in counting:

$$\frac{N}{A} = 2 \cdot \frac{A_{\text{FS}}}{(2\pi)^2}.$$

Use $A_{\text{FS}} = A_{\text{BZ}}/100 = (2\pi/a)^2/100$:

$$\frac{N}{A} = \frac{2}{100} \cdot \frac{(2\pi/a)^2}{(2\pi)^2}.$$

Cancel $(2\pi)^2$:

$$\frac{N}{A} = \frac{1}{50 a^2}.$$

Plug in $a = 3.5 \times 10^{-10} \text{ m}$:

$$n_{2D} = \frac{1}{50 \times (3.5 \times 10^{-10})^2} \text{ m}^{-2}.$$

Reduce $50 \times (3.5)^2 = 612.5$:

$$n_{2D} = \frac{1}{6.125 \times 10^{-18}} \text{ m}^{-2}.$$

$$\boxed{n_{2D} = 1.63 \times 10^{17} \text{ m}^{-2}.} \quad (29)$$

(c)(ii) Hall voltage

2D Hall coefficient (single carrier, magnitude):

$$R_H^{(2D)} = \frac{1}{n_{2D} e}.$$

Hall voltage across the strip equals $E_y \times w$ but the 2D current density is per unit width:

$$V_H = \frac{j_x B_z}{n_{2D} e}.$$

Plug in $j_x = 10 \mu\text{A m}^{-1} = 1.0 \times 10^{-5} \text{ A m}^{-1}$ and $B_z = 1 \text{ T}$:

$$V_H = \frac{1.0 \times 10^{-5} \times 1}{1.63 \times 10^{17} \times 1.602 \times 10^{-19}} \text{ V}.$$

Evaluate the denominator:

$$1.63 \times 10^{17} \times 1.602 \times 10^{-19} = 2.612 \times 10^{-2}.$$

Reduce:

$$V_H = \frac{1.0 \times 10^{-5}}{2.612 \times 10^{-2}} \text{ V} = 3.83 \times 10^{-4} \text{ V}.$$

$$\boxed{V_H \approx 3.8 \times 10^{-4} \text{ V} \approx 0.38 \text{ mV}.} \quad (30)$$

Question 4 — Magnetism: Hund's rules, mean-field BCC ferromagnet

Problem. (a) Hund's rules for Mn^{2+} ($3d^5$), Tb^{3+} ($4f^8$); $T = 0$ moments. (b) $S = 1/2$ Heisenberg + Zeeman on bcc (edge a): mean-field B_{mf} , self-consistent M , Curie-Weiss χ , $\Theta = T_C$, find J_1 for $T_C = 30 \text{ K}$. (c) Same on simple cubic with $T'_C = 20 \text{ K}$; find J'_1 ; sketch $\chi^{-1}(T)$.

(a)(i) Hund's rules for Mn^{2+} ($3d^5$) and Tb^{3+} ($4f^8$)

Mn^{2+} , $3d^5$. Five d electrons, one in each $m_\ell \in \{-2, -1, 0, 1, 2\}$, all spin up:

$$S = 5 \times \frac{1}{2} = \frac{5}{2}.$$

Sum the m_ℓ values:

$$L = -2 - 1 + 0 + 1 + 2 = 0.$$

Half-filled shell with $L = 0$ gives $J = S$:

$$J = \frac{5}{2}.$$

Apply the Landé formula with $S(S+1) = 35/4$, $L(L+1) = 0$, $J(J+1) = 35/4$:

$$g_J = \frac{3}{2} + \frac{35/4 - 0}{2 \times 35/4}.$$

Reduce:

$$g_J = \frac{3}{2} + \frac{1}{2} = 2.$$

$$\boxed{\text{Mn}^{2+} : S = \frac{5}{2}, L = 0, J = \frac{5}{2}, g_J = 2.} \quad (31)$$

Tb^{3+} , $4f^8$. Place 7 electrons one in each $m_\ell \in \{-3, \dots, 3\}$ then the 8th pairs into $m_\ell = +3$:

$$S = \frac{7}{2} - \frac{1}{2} = 3.$$

Sum the m_ℓ values: the 7 aligned electrons give $L = 0$, the extra spin-down at $m_\ell = +3$ adds:

$$L = 0 + 3 = 3.$$

More than half-full shell gives $J = L + S$:

$$J = 3 + 3 = 6.$$

Apply Landé with $S(S+1) = 12$, $L(L+1) = 12$, $J(J+1) = 42$:

$$g_J = \frac{3}{2} + \frac{12 - 12}{2 \times 42}.$$

Reduce:

$$g_J = \frac{3}{2} + 0 = \frac{3}{2}.$$

$$\boxed{\text{Tb}^{3+} : S = 3, L = 3, J = 6, g_J = \frac{3}{2}.} \quad (32)$$

(a)(ii) Saturation magnetic moment at $T = 0$

At $T = 0$ the moment is $m = g_J J \mu_B$. Mn^{2+} :

$$m_{\text{Mn}^{2+}} = 2 \times \frac{5}{2} \mu_B = 5 \mu_B.$$

Tb^{3+} :

$$m_{\text{Tb}^{3+}} = \frac{3}{2} \times 6 \mu_B = 9 \mu_B.$$

$$\boxed{m_{\text{Mn}^{2+}} = 5 \mu_B, \quad m_{\text{Tb}^{3+}} = 9 \mu_B.} \quad (33)$$

(b)(i) Mean-field effective field on a bcc site

bcc coordination $z = 8$. Replace $\mathbf{S}_j \rightarrow \langle \mathbf{S} \rangle$ in the Heisenberg sum acting on site i :

$$\mathcal{H}_i^{\text{MF}} = -J_1 z \langle \mathbf{S} \rangle \cdot \mathbf{S}_i + g\mu_B \mathbf{B} \cdot \mathbf{S}_i. \quad (34)$$

Define $\mathbf{B}_{\text{mf}}$ by $\mathcal{H}_i^{\text{MF}} \equiv g\mu_B \mathbf{B}_{\text{mf}} \cdot \mathbf{S}_i$:

$$\mathbf{B}_{\text{mf}} = \mathbf{B} - \frac{J_1 z}{g\mu_B} \langle \mathbf{S} \rangle.$$

Magnetisation in terms of $\langle \mathbf{S} \rangle$ uses $\boldsymbol{\mu} = -g\mu_B \mathbf{S}$:

$$\mathbf{M} = -ng\mu_B \langle \mathbf{S} \rangle.$$

Solve for $\langle \mathbf{S} \rangle$:

$$\langle \mathbf{S} \rangle = -\frac{\mathbf{M}}{ng\mu_B}.$$

Substitute into $\mathbf{B}_{\text{mf}}$:

$$\mathbf{B}_{\text{mf}} = \mathbf{B} + \frac{J_1 z}{(g\mu_B)^2 n} \mathbf{M}.$$

Identify λ via $\mathbf{B}_{\text{mf}} = \mathbf{B} + \lambda \mathbf{M}$:

$$\lambda = \frac{J_1 z}{(g\mu_B)^2 n}.$$

Insert $z = 8$ and $n = 2/a^3$:

$$\lambda = \frac{8J_1}{(g\mu_B)^2 \cdot 2/a^3}.$$

Simplify:

$$\boxed{\lambda = \frac{4J_1 a^3}{(g\mu_B)^2}}. \quad (35)$$

(b)(ii) Self-consistent magnetisation

Single $S = 1/2$ in B_{mf} has partition function:

$$Z = 2 \cosh\left(\frac{g\mu_B B_{\text{mf}}}{2k_B T}\right).$$

Spin expectation along the field:

$$\langle S_z \rangle = -\frac{1}{2} \tanh\left(\frac{g\mu_B B_{\text{mf}}}{2k_B T}\right).$$

Magnetisation $M = -ng\mu_B \langle S_z \rangle$:

$$M = \frac{ng\mu_B}{2} \tanh\left(\frac{g\mu_B B_{\text{mf}}}{2k_B T}\right).$$

Identify $M_0 = ng\mu_B/2$:

$$M = M_0 \tanh\left(\frac{g\mu_B B_{\text{mf}}}{2k_B T}\right).$$

Insert $n = 2/a^3$:

$$M_0 = \frac{2}{a^3} \cdot \frac{g\mu_B}{2}.$$

Simplify:

$$\boxed{M_0 = \frac{g\mu_B}{a^3}}. \quad (36)$$

(b)(iii) Curie–Weiss susceptibility above T_C

Expand $\tanh x \approx x$ for small argument, with $B_{\text{mf}} = B + \lambda M$:

$$M \simeq M_0 \cdot \frac{g\mu_B(B + \lambda M)}{2k_B T}.$$

Collect terms linear in M :

$$M \left(1 - \frac{M_0 g\mu_B \lambda}{2k_B T} \right) = \frac{M_0 g\mu_B}{2k_B T} B.$$

Divide and identify $\chi = M/B$:

$$\chi = \frac{M_0 g\mu_B / (2k_B T)}{1 - M_0 g\mu_B \lambda / (2k_B T)}.$$

Multiply top and bottom by T :

$$\chi = \frac{M_0 g\mu_B / (2k_B)}{T - M_0 g\mu_B \lambda / (2k_B)} \equiv \frac{C}{T - \Theta}.$$

Compute Θ with $M_0 = g\mu_B/a^3$ and $\lambda = 4J_1 a^3 / (g\mu_B)^2$:

$$\Theta = \frac{g\mu_B}{a^3} \cdot g\mu_B \cdot \frac{4J_1 a^3}{(g\mu_B)^2} \cdot \frac{1}{2k_B}.$$

Cancel $(g\mu_B)^2$ and a^3 :

$$\Theta = \frac{4J_1}{2k_B} = \frac{2J_1}{k_B}.$$

Compute C :

$$C = \frac{M_0 g\mu_B}{2k_B} = \frac{g\mu_B}{a^3} \cdot \frac{g\mu_B}{2k_B}.$$

Simplify:

$$C = \frac{(g\mu_B)^2}{2k_B a^3}.$$

$$\boxed{\Theta = \frac{2J_1}{k_B}, \quad C = \frac{(g\mu_B)^2}{2k_B a^3}.} \quad (37)$$

(b)(iv) Relation $\Theta = T_C$

At $B = 0$ the self-consistent equation reads:

$$M = M_0 \tanh \left(\frac{g\mu_B \lambda M}{2k_B T} \right).$$

A non-zero M first appears when the slope at $M = 0$ equals 1:

$$\frac{M_0 g\mu_B \lambda}{2k_B T_C} = 1.$$

Solve for T_C :

$$T_C = \frac{M_0 g\mu_B \lambda}{2k_B} = \Theta.$$

$$\boxed{T_C = \Theta = \frac{2J_1}{k_B}.} \quad (38)$$

(b)(v) Numerical J_1 for $T_C = 30$ K

Solve $T_C = 2J_1/k_B$ for J_1 :

$$J_1 = \frac{k_B T_C}{2}.$$

Plug in $T_C = 30$ K:

$$J_1 = \frac{1.381 \times 10^{-23} \times 30}{2} \text{ J}.$$

Reduce:

$$J_1 = 2.072 \times 10^{-22} \text{ J}.$$

Convert to meV via $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$:

$$J_1 = \frac{2.072 \times 10^{-22}}{1.602 \times 10^{-19}} \text{ eV} = 1.29 \text{ meV}.$$

$$J_1 \approx 2.07 \times 10^{-22} \text{ J} \approx 1.3 \text{ meV}, \quad J_1/k_B \approx 15 \text{ K}.$$

(39)

(c)(i) J'_1 for the simple-cubic lattice with $T'_C = 20$ K

Simple cubic: $z' = 6$, $n' = 1/a^3$. Repeat (b)(i):

$$\lambda' = \frac{J'_1 z'}{(g\mu_B)^2 n'} = \frac{6J'_1}{(g\mu_B)^2 \cdot 1/a^3}.$$

Simplify:

$$\lambda' = \frac{6J'_1 a^3}{(g\mu_B)^2}.$$

Repeat (b)(ii) with $n' = 1/a^3$:

$$M'_0 = \frac{n' g\mu_B}{2} = \frac{g\mu_B}{2a^3}.$$

Compute Θ' :

$$\Theta' = \frac{M'_0 g\mu_B \lambda'}{2k_B}.$$

Substitute:

$$\Theta' = \frac{g\mu_B}{2a^3} \cdot g\mu_B \cdot \frac{6J'_1 a^3}{(g\mu_B)^2} \cdot \frac{1}{2k_B}.$$

Cancel $(g\mu_B)^2$ and a^3 :

$$\Theta' = \frac{6J'_1}{4k_B} = \frac{3J'_1}{2k_B}.$$

Set $T'_C = \Theta'$ and solve:

$$J'_1 = \frac{2k_B T'_C}{3}.$$

Plug in $T'_C = 20$ K:

$$J'_1 = \frac{2 \times 1.381 \times 10^{-23} \times 20}{3} \text{ J}.$$

Reduce:

$$J'_1 = \frac{5.524 \times 10^{-22}}{3} \text{ J} = 1.841 \times 10^{-22} \text{ J}.$$

Convert to meV:

$$J'_1 = \frac{1.841 \times 10^{-22}}{1.602 \times 10^{-19}} \text{ eV} = 1.15 \text{ meV}.$$

$$J'_1 \approx 1.84 \times 10^{-22} \text{ J} \approx 1.15 \text{ meV}, \quad J'_1/k_B \approx 13.3 \text{ K}.$$

(40)

(c)(ii) Sketch of $\chi^{-1}(T)$ for both materials

$\chi^{-1} = (T - \Theta)/C$, slope $1/C$, T -intercept Θ . With $C_{\text{bcc}} = (g\mu_{\text{B}})^2/(2k_{\text{B}}a^3)$ and $C_{\text{sc}} = (g\mu_{\text{B}})^2/(4k_{\text{B}}a^3)$, the sc slope is twice the bcc slope.

